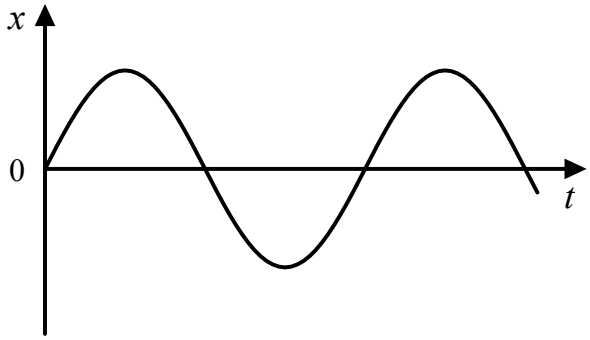
STPM 2015 PHYSICS 960/3 (U3)

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg, $g = 9.81$ m s^{-2}.

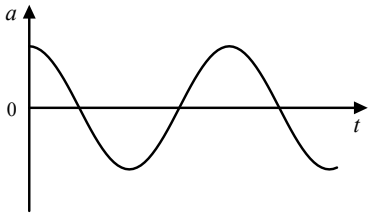
Section A [15 marks]

1. The displacement x of an object varies with time t when it oscillates with a simple harmonic motion is shown in the graph below.

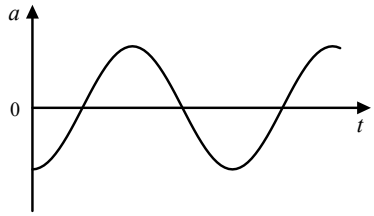


Which graph shows the acceleration a of the object varies with time t?

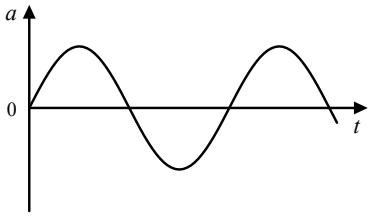
A



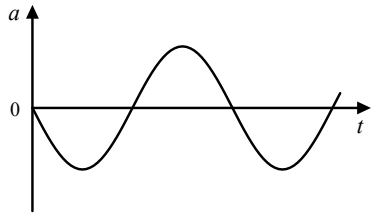
B



C



D



Answer: D

$a = -\omega^2 x$: the acceleration graph is the displacement graph inverted

x starts at zero and increases, so a starts at zero and becomes negative (D).

Explanation: Acceleration and displacement are always in antiphase in SHM.

2. A mass of 25 g is hung at the lower end of a vertical helical spring with a force constant of 2.8 N m^{-1}. The mass is pulled down then released to undergo vertical oscillations. What is the additional mass required so that the period of the vertical oscillations is 0.70 s?

A 4.5 g B 9.8 g C 30 g D 35 g

Answer: B

$$T = 2\pi\sqrt{(\frac{m}{k})} \Longrightarrow m = k(\frac{T}{2\pi})^2 = 2.8(\frac{0.70}{2\pi})^2 = 34.8 \text{ g}$$

Additional mass = 34.8 − 25 = 9.8 g

Explanation: The period of a mass–spring system depends only on the mass and the spring constant.

3. A point source produces spherical waves. The power emitted by the source is 0.80 W. What is the intensity of the waves at a distance of 2.0 m from the source?

A 0.016 W m^{-2} B 0.064 W m^{-2} C 0.40 W m^{-2} D 40 W m^{-2}

Answer: A

$$I = \frac{P}{4\pi r^2} = \frac{0.80}{4\pi(2.0)^2} = 0.016 \text{ W m}^{-2}$$

Explanation: The power spreads over the surface of a sphere.

4. Which statement is **not** true about mechanical and electromagnetic waves?

A Mechanical and electromagnetic waves can undergo polarisation.
B Mechanical and electromagnetic waves can undergo interference and diffraction.
C Mechanical waves can be longitudinal or transverse waves but all electromagnetic waves are transverse waves.
D Mechanical waves require a medium for propagation but electromagnetic waves do not require a medium for propagation.

Answer: A

Longitudinal mechanical waves (e.g. sound) cannot be polarised, so statement A is not true.

Explanation: All waves show interference and diffraction; only transverse waves can be polarised.

5. Ali and Baba attend a musical concert. Ali brings a noise dosimeter and measures a sound level of 108 dB from his seat. Ali and Baba sit at 10.0 m and 5.0 m respectively from the speaker. What is the intensity level of the sound from Baba's seat?

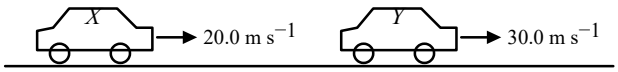
A 108 dB B 111 dB C 114 dB D 118 dB

Answer: C

Halving the distance makes the intensity 4 times: $\beta = 108 + 10 \log 4 = 114$ dB

Explanation: Inverse square law: $I \propto 1/r^2$; each doubling of intensity adds 3 dB.

6. Cars *X* and *Y* are moving in the same direction with their respective velocity shown in the diagram below.



If the driver of car *X* honks the horn with a frequency of 1000 Hz, what is the frequency of the sound heard by the driver of car *Y*?
[The speed of sound in air is 300 m s^{-1}.]

A 844 Hz B 964 Hz C 1030 Hz D 1180 Hz

Answer: B

X (source) behind, moving towards *Y*; *Y* (observer) moving away: $f' = 1000\dfrac{300 - 30.0}{300 - 20.0} = 964$ Hz

Explanation: *Y* moves away faster than *X* follows, so the separation increases and the frequency heard is lower.

7. A converging lens produces an inverted image which is twice the size of the object when the object is at a distance of 30 cm from the lens. What is the new distance of the object from the lens if the size of the newly formed image is also twice the size of the object?

A 5.0 cm B 10.0 cm C 15.0 cm D 20.0 cm

Answer: B

$$\text{Real image, } m = 2\text{: } v = 60 \text{ cm} \Longrightarrow \frac{1}{f} = \frac{1}{30} + \frac{1}{60} \Longrightarrow f = 20 \text{ cm}$$

$$\text{Virtual image, } m = 2\text{: } v = -2u \Longrightarrow \frac{1}{u} - \frac{1}{2u} = \frac{1}{20} \Longrightarrow u = 10.0 \text{ cm}$$

Explanation: A converging lens gives a twice-size image in two positions: real (beyond *f*) and virtual (inside *f*).

8. When monochromatic light is incident normally on a narrow slit, a diffraction pattern is formed on the screen. Which statement is true of the diffraction pattern if the slit is narrower?

A The central bright fringe is broadened.
B The central bright fringe becomes a dark fringe.
C The intensity of each fringe pattern is the same.
D The intensity of the diffraction pattern increases.

Answer: A

Width of central fringe = $\dfrac{2\lambda D}{a}$: a narrower slit gives a broader central bright fringe

Explanation: Less light passes through a narrower slit, so the pattern also becomes dimmer.

9. Parallel light of wavelength 590 nm is incident normally on a diffraction grating. The first order maximum is observed at 30.0° to the normal. What is its grating separation?

A 295 nm B 590 nm C 1180 nm D 2360 nm

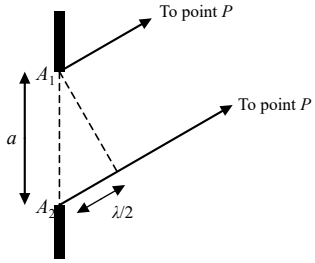
Answer: C

$$d \sin\theta = \lambda \Longrightarrow d = \frac{590}{\sin 30.0°} = 1180 \text{ nm}$$

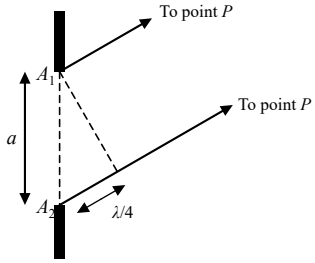
Explanation: First order, $n = 1$.

10. Monochromatic light is incident on a single slit of width a. Secondary sources, A_1 and A_2, emit wavelets that meet at point P on a screen. Which diagram shows the correct path difference between the two wavelets to produce the first minimum of the diffraction pattern at point P?

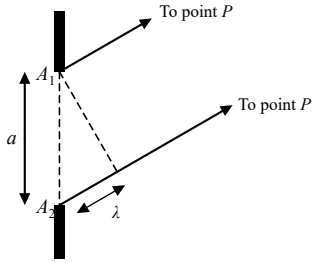
A



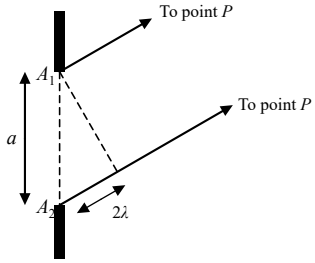
B



C



D



Answer: C

First minimum: the path difference between the wavelets from the two edges of the slit is λ (diagram C).

Explanation: Then the slit can be divided into two halves whose corresponding wavelets differ by $\lambda/2$ and cancel in pairs.

11. If α-particle and proton are moving with the same de Broglie wavelength, which statement is true about the particles?

A The speed of the α-particle is greater than that of the proton.
B The speed of the α-particle is the same as that of the proton.
C The momentum of the α-particle is the same as that of the proton.
D The momentum of the α-particle is greater than that of the proton.

Answer: C

$\lambda = \frac{h}{p}$: equal wavelengths $\Longrightarrow$ equal momenta

Explanation: The α particle is heavier, so with the same momentum it moves more slowly.

12. What is the de Broglie wavelength of a proton with a kinetic energy of 100 eV?

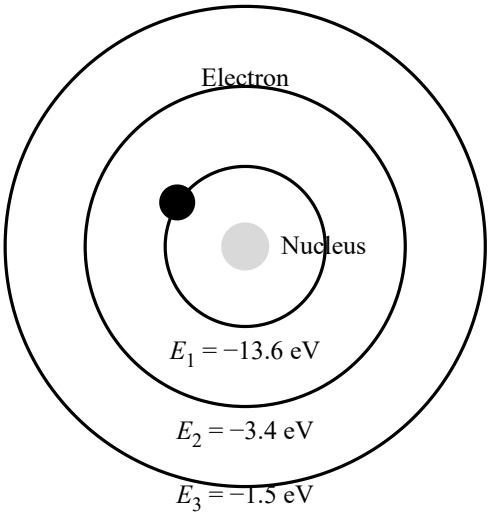
A 4.14×10^{-17} m B 2.87×10^{-12} m C 5.74×10^{-12} m D 1.23×10^{-10} m

Answer: B

$$\lambda = \frac{h}{\sqrt{(2mK)}} = \frac{6.63 \times 10^{-34}}{\sqrt{(2(1.67 \times 10^{-27})(100)(1.60 \times 10^{-19}))}} = 2.87 \times 10^{-12} \text{ m}$$

Explanation: Convert 100 eV to joules.

13. A hydrogen atom can undergo a transition state when collision occurs between the hydrogen atom and electron or when it interacts with a photon. Energy levels for the three states in the hydrogen atom are shown in the diagram below.



Which statement is **not** true about the transition?

A When the energy of incident electron is 11.0 eV, the electron will be excited from E_1 to E_2.

B When the energy of incident electron is 13.0 eV, the electron will be excited from E_1 to E_3.

C When the energy of incident photon is 10.2 eV, the electron will be excited from E_1 to E_2.

D When the energy of incident photon is 11.0 eV, the electron will be excited from E_1 to E_3.

Answer: D

A photon is absorbed only if its energy exactly equals a level difference. $E_3 - E_1 = 12.1$ eV, so an 11.0 eV photon cannot excite $E_1 \rightarrow E_3$. Statement D is not true.

Explanation: An incident electron can give up part of its energy, so electrons with energy above 10.2 eV (A) or 12.1 eV (B) can cause these excitations.

14. The mass of a beryllium nucleus $^{7}_{4}$Be is 7.0147 u. What is the binding energy per nucleon of this nucleus?
[Mass of a proton = 1.0073 u, mass of a neutron = 1.0087 u; 1 u ≡ 932 MeV.]

A 5.41 MeV B 5.59 MeV C 37.8 MeV D 39.14 MeV

Answer: A

$$\Delta m = 4(1.0073) + 3(1.0087) - 7.0147 = 0.0406 \text{ u}$$

$$E_B/A = \frac{(0.0406)(932)}{7} = 5.41 \text{ MeV}$$

Explanation: Be-7 has 4 protons and 3 neutrons.

15. The half-life of a radioactive sample is 1600 years. How long would it take $\frac{7}{8}$ of the sample to decay?

A 200 years B 1400 years C 2800 years D 4800 years

Answer: D

$\frac{7}{8}$ decayed $\Longrightarrow$ $\frac{1}{8} = (\frac{1}{2})^3$ remains $\Longrightarrow$ 3 half-lives = 3 × 1600 = 4800 years

Explanation: Work from the fraction remaining, not the fraction decayed.

Section B [15 marks]

16. (a) State **one** characteristic of a simple harmonic motion. [1]

Answer

The acceleration is directly proportional to the displacement from the equilibrium position and always directed towards it ($a = -\omega^2 x$)

1

1

Explanation: Other acceptable answers: constant period independent of amplitude; total energy constant.

(b) The oscillation of a simple pendulum is given as $x = 2.4 \sin 3.0t$, where x is the displacement in centimeters and t is the time in seconds.

(i) Determine the frequency of the oscillation. [3]

Answer

Compare with $x = A \sin \omega t$: $\omega = 3.0$ rad s^{-1} 1

$f = \frac{\omega}{2\pi} = \frac{3.0}{2\pi}$ 1

$f = 0.477$ Hz 1

3

Explanation: The angular frequency is the coefficient of t.

(ii) Calculate the speed of the pendulum bob at time $t = 0.40$ s. [2]

Answer

$v = \frac{dx}{dt} = (2.4)(3.0) \cos (3.0 \times 0.40) = 7.2 \cos 1.2$ rad 1

$v = 2.61$ cm s^{-1} 1

2

Explanation: Use radians.

(iii) Determine the length of the simple pendulum. [2]

Answer

$\omega^2 = \frac{g}{l} \Rightarrow l = \frac{9.81}{3.0^2}$ 1

$l = 1.09$ m 1

2

Total for question **8**

Explanation: For a simple pendulum $\omega = \sqrt{(\frac{g}{l})}$.

17. (a)

(i) Describe a nuclear chain reaction. [2]

Answer

A neutron causes a heavy nucleus (e.g. U-235) to split, releasing energy and two or three neutrons 1

These neutrons go on to cause further fissions, so the reaction becomes self-sustaining 1

2

Explanation: Each fission releases about 200 MeV.

(ii) How is the chain reaction controlled in a nuclear reactor? [2]

Answer

Control rods (e.g. boron or cadmium) are inserted into the core to absorb excess neutrons	1
They are adjusted so that on average one neutron from each fission causes another fission (a steady rate)	1
	2

Explanation: A moderator slows the neutrons so that they can cause fission efficiently.

(b) Determine a and X for the following nuclear reaction.
$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + aX$ [3]

Answer

Nucleon number: $235 + 1 = 144 + 89 +$ (nucleon number of aX) $\Longrightarrow 3$	1
Proton number: $92 = 56 + 36 +$ (proton number of aX) $\Longrightarrow 0$	1
X is a neutron, ${}^{1}_{0}\text{n}$, and $a = 3$	1
	3
Total for question	**7**

Explanation: Particles with nucleon number 1 and proton number 0 are neutrons.

Section C [30 marks]

18. (a) State **two** differences between progressive and standing waves. [2]

Answer

• A progressive wave transfers energy; a standing wave does not	1
• In a progressive wave all points have the same amplitude; in a standing wave the amplitude varies from zero (nodes) to maximum (antinodes)	1
	2

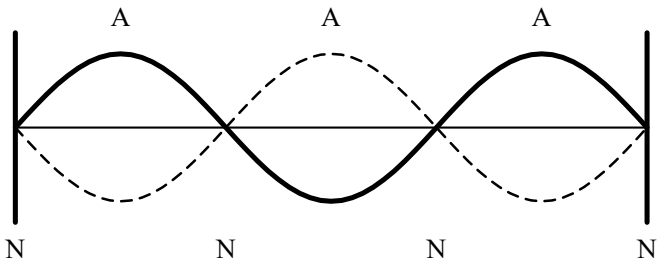
Explanation: Also: the waveform of a progressive wave moves; that of a standing wave does not.

(b) Two waves, P and Q, travel in opposite directions along a stretched string are represented by $y_P = 0.015 \sin (2.0t - 8.0x)$ and $y_Q = 0.015 \sin (2.0t + 8.0x)$, where x and y are in meters and t is

in seconds. These two waves superposed one another to form a standing wave with a second overtone.

(i) Sketch the pattern of the standing wave for the second overtone. [1]

Answer



Three loops with nodes at both ends 1

1

Explanation: The second overtone is the third harmonic.

(ii) Derive an equation of the standing wave formed. [3]

Answer

$y = y_P + y_Q = 0.015[\sin(2.0t - 8.0x) + \sin(2.0t + 8.0x)]$ 1

$= 2(0.015)\sin(2.0t)\cos(8.0x)$ 1

$y = 0.030\cos(8.0x)\sin(2.0t)$ m 1

3

Explanation: Use $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$.

(iii) Calculate the amplitude of the standing wave at $x = 0.12$ m. [2]

Answer

Amplitude $= 0.030|\cos(8.0 \times 0.12)| = 0.030|\cos 0.96 \text{ rad}|$ 1

$= 0.0172$ m 1

2

Explanation: Use radians.

(iv) Determine the distance between two consecutive nodes of the standing wave. [4]

Answer

Nodes where $\cos(8.0x) = 0$	1
$8.0x = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$	1
Distance between consecutive nodes $= \frac{\pi}{8.0}$	1
$= 0.393$ m	1
	4

Explanation: This is half a wavelength ($\lambda = 2\pi/8.0 = 0.785$ m).

(v) Determine **two** possible values of x for the positions of antinodes of the standing wave. [3]

Answer

Antinodes where $\lvert\cos(8.0x)\rvert = 1$: $8.0x = 0, \pi, 2\pi, \ldots$	1
$x = 0, \frac{\pi}{8}, \ldots$	1
e.g. $x = 0.393$ m and 0.785 m	1
	3
Total for question	**15**

Explanation: Antinodes are midway between nodes.

19. (a) An object of height 5.0 mm is placed 35.0 cm from the vertex of a spherical surface of a glass rod. An image is formed inside the rod 50.0 cm from the vertex of the glass rod. The refractive index of the glass is 1.5.

(i) State **two** characteristics of the image. [2]

Answer

Real (formed by actual rays inside the rod)	1
Inverted (and magnified)	1
	2

Explanation: A positive image distance inside the rod means a real image.

(ii) Determine the radius of curvature of the spherical surface of the glass rod. [3]

Answer

$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$ 1

$\frac{1}{35.0} + \frac{1.5}{50.0} = \frac{0.5}{r}$ 1

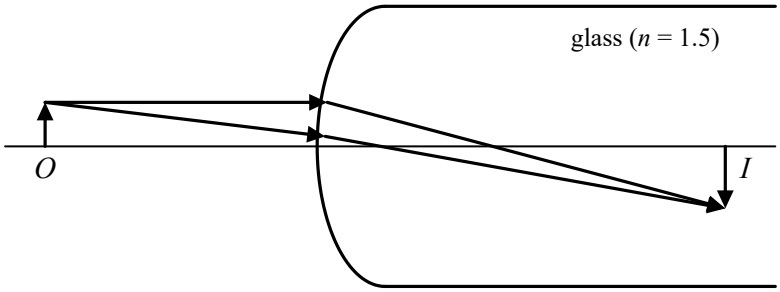
$r = 8.54$ cm (convex surface) 1

3

Explanation: Light travels from air ($n_1 = 1$) into glass ($n_2 = 1.5$).

(iii) Sketch a ray diagram to show the formation of the image. [2]

Answer



Rays from the object refracted at the curved surface 1

converging to form a real, inverted image inside the rod 1

2

Explanation: Rays bend towards the normal on entering the glass.

(iv) Determine the position of the object if the virtual image is formed 10.0 cm from the vertex of the glass rod. [2]

Answer

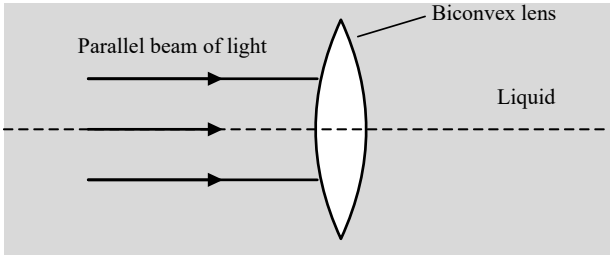
$\frac{1}{u} + \frac{1.5}{-10.0} = \frac{0.5}{8.54}$ 1

$u = 4.80$ cm from the vertex 1

2

Explanation: A virtual image forms when the object is close to the surface.

(b) A thin biconvex lens with radii of curvature 10.0 cm which is immersed in a liquid of refractive index 1.65 is shown in the diagram below.



The refractive index of the lens is 1.5.

(i) Determine the focal length of the lens. [3]

Answer

$\frac{1}{f} = (\frac{n_{\text{lens}}}{n_{\text{liquid}}} - 1)(\frac{1}{r_1} + \frac{1}{r_2})$ 1

$\frac{1}{f} = (\frac{1.5}{1.65} - 1)(\frac{1}{10.0} + \frac{1}{10.0})$ 1

$f = -55$ cm 1

3

Explanation: Use the refractive index of the lens relative to the surrounding liquid.

(ii) State the type of the lens. [1]

Answer

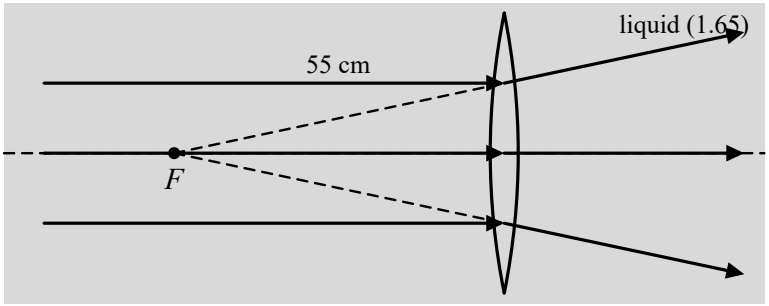
Diverging lens (negative focal length) 1

1

Explanation: The lens is optically less dense than the liquid, so a biconvex shape diverges light.

(iii) Sketch a ray diagram to show the position of the focal point of the lens. [2]

Answer



Parallel rays diverge after the lens	1
Backward extensions meet at a virtual focal point 55 cm in front of the lens	1
	2
Total for question	**15**

Explanation: This behaves like a concave lens in air.

20. (a) State Bohr's postulates. [2]

Answer

Electrons move in certain allowed circular orbits without radiating energy; the angular momentum is quantised, $mvr = \frac{nh}{2\pi}$	1
A photon is emitted or absorbed only when an electron jumps between orbits: $hf = E_i - E_f$	1
	2

Explanation: These postulates explain the line spectrum of hydrogen.

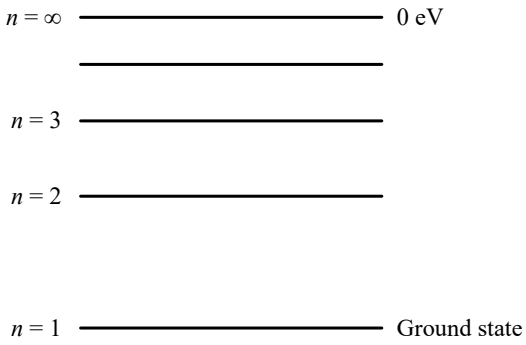
(b) Derive an expression for the orbital radii of the atom in Bohr's model. [3]

Answer

$\frac{mv^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2}$ and $mvr = \frac{nh}{2\pi}$	1
Eliminate $v = \frac{nh}{2\pi mr}$: $\frac{n^2h^2}{4\pi^2 mr} = \frac{e^2}{4\pi\varepsilon_0}$	1
$r_n = \frac{\varepsilon_0 h^2 n^2}{\pi m e^2}$	1
	3

Explanation: The radii are proportional to n^2.

(c) The energy levels of a hypothetical hydrogen-like atom are shown in the diagram below.



(i) What is meant by *ionisation energy*? [2]

Answer

The minimum energy needed to remove an electron completely from the atom in its ground state — 1

i.e. to move it from $n = 1$ to $n = \infty$ — 1

2

Explanation: It equals the magnitude of the ground-state energy.

(ii) During the transition of an atom from level 3 to level 2, a photon of wavelength 73.13 nm is emitted. Calculate the ionisation energy for this atom. Express your answer in electron volt. [5]

Answer

Photon energy $= \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{73.13 \times 10^{-9}} = 2.72 \times 10^{-18}$ J $= 17.0$ eV — 1

Hydrogen-like levels: $E_n = -\frac{E_{\text{ion}}}{n^2}$ — 1

$E_3 \rightarrow E_2$: $E_{\text{ion}}(\frac{1}{4} - \frac{1}{9}) = 17.0$ eV — 1

$E_{\text{ion}} = \frac{17.0}{5/36}$ — 1

$E_{\text{ion}} = 122$ eV (122.4 eV) — 1

5

Explanation: 122.4 eV = 9 × 13.6 eV, the ionisation energy of a hydrogen-like ion with $Z = 3$ (Li^{2+}).

(d) Why will certain minerals glow under ultraviolet light? [3]

Answer

The minerals absorb ultraviolet photons, exciting electrons to high energy levels	1
The electrons return to lower levels in several smaller steps	1
emitting photons of lower energy (longer wavelength) in the visible region, so the minerals glow (fluorescence)	1
	3
Total for question	**15**

Explanation: The emitted visible light has less energy per photon than the absorbed ultraviolet.